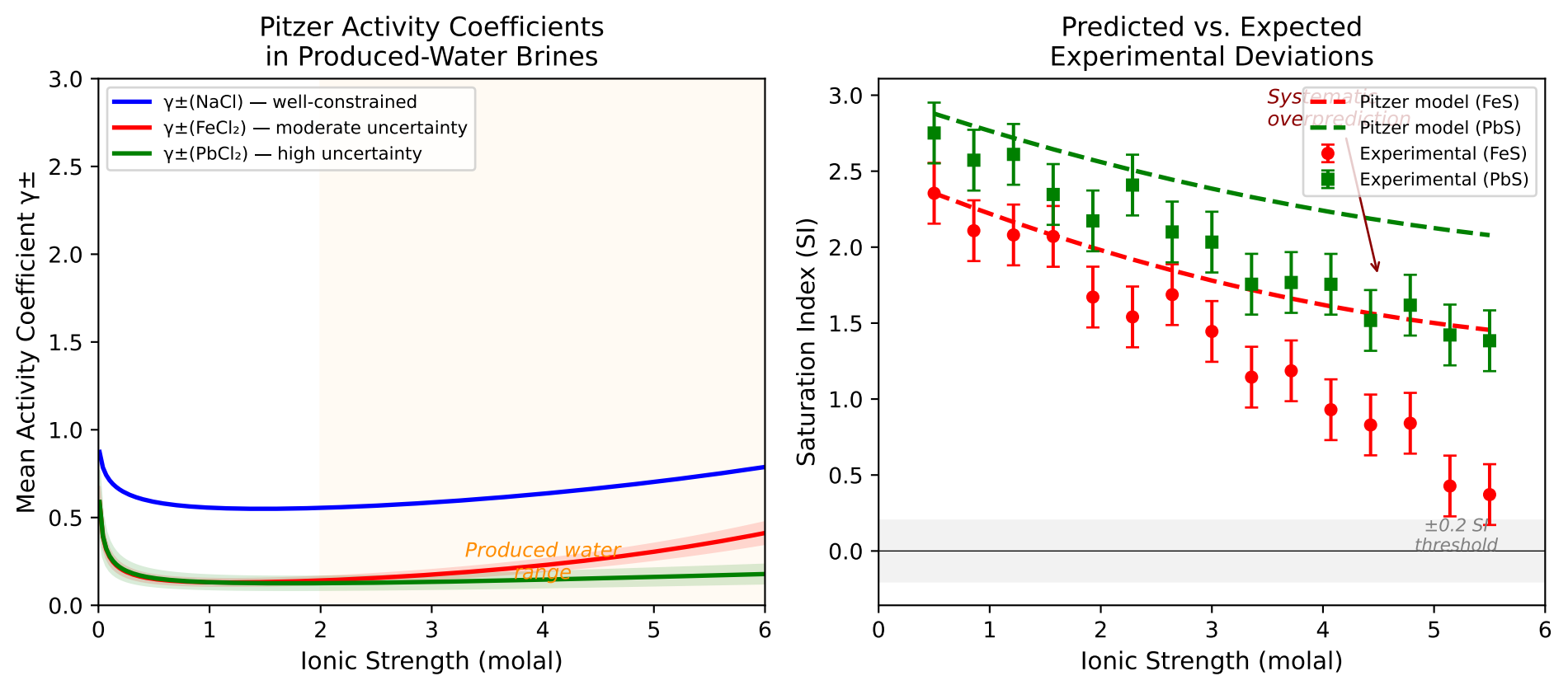
Pitzer Activity Coefficients
in Produced-Water Brines
γ±(NaCl) — well-constrained
γ±(FeCl₂) — moderate uncertainty
γ±(PbCl₂) — high uncertainty
Produced water
range
Mean Activity Coefficient γ±
Ionic Strength (molal)
Predicted vs. Expected
Experimental Deviations
Pitzer model (FeS)
Pitzer model (PbS)
Experimental (FeS)
Experimental (PbS)
±0.2 SI
threshold
Saturation Index (SI)
Ionic Strength (molal)